

In fact, almost any smart device in your home that uploads data to a third party server can be used to spy on you. A smart electricity meter could be hacked to identify electronic signatures unique to a particular programme to guess the shows you're watching. One doesn't even need to go that far if you're using a smart television, especially one that has been dropped from the regular update cycle. Something like the Weeping Angel exploit which has been used by CIA could be used by anyone close to you who can plug a USB into your Samsung smart TV. And don't even get us started on what is possible by hijacking your smartphone's or your laptop's camera and microphone. Vulnerabilities being uncovered everyday could reveal

such exploits existing for quite a while and open new doors.

Apart from the new ways in which existing technology can be used to spy on you, expect some brand new tech to show up in the fray as well. While the Government of India has not entirely been clear about drones, it has shown, through action, that drones are not entirely welcome en masse. But what we're talking about are a different breed of drones, ones so small and silent that you won't be able to detect their presence unless they're up close. And with facial recognition technology becoming really accessible, it won't be long before a hobbyist drone could replace the local stalker. A good mix of both new and old tech is

going to happen in the world of CCTV surveillance. If you're thinking that it's going to be a while before police can upload someone's mugshot to a surveillance network and tail them around a city, or a suspicious mall security network flags certain individuals to law enforcement pre-emptively, you're about to be surprised. IC Realtime's flagship product Ella, an app and web-based AI platform for making video feeds searchable, was unveiled last December. The platform can pick up all kinds of natural language queries - for instance, looking for people wearing certain clothes and using certain cars.

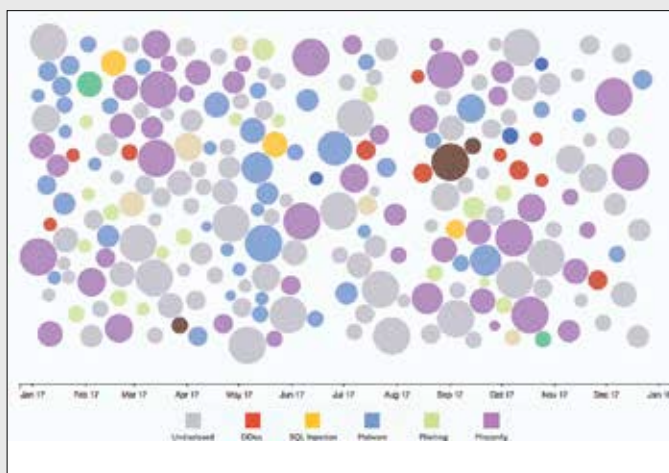
To be safe, make sure that all devices that are connected to the internet are updated with security patches. ■

BIG DATA BREACHES

Over the course of 2017, over 99 billion records were compromised because of data breaches. The figure is actually much higher, as in 60 percent of the breaches, the number of compromised records are not known or not reported. Expect at least a few major breaches in 2018. Companies may not even be aware that a data breach has occurred, or of a security flaw in their systems, so expect revelations of data breaches that have already occurred, to be publicly revealed in 2018.

As companies move to increasingly store their data on the cloud, expect major cloud data breaches in 2018. A part of the problem is that security professionals do not yet fully understand the practices needed to protect cloud data. Security researchers anticipate that bad configurations for securing cloud storage will be a major reason behind data breaches. In 2017, the records of 14 million Verizon customers was found on an unprotected Amazon S3 server, 100 gigabytes of NSA data on a virtual disk was leaked, and a vulnerability with Cloudflare's edge servers allowed for the Uber breach.

There are legal measures being put into place when it comes to disclosing the breach to the affected parties, as well as have better approaches in place to address the situation and mitigate the damage. This is because the General Data Protection Regulation (GDPR) in Europe will come into effect start-



Credit: IBM X-Force

Data breaches by attack method over the past year. Notice that in most of the cases, the method is undisclosed.

ing May 2018. The GDPR requires organisations to report data breaches to regulators within 72 hours of discovering a breach. Countries around the world are expected to enforce similar regulations. In India too, the Ministry of Electronics and Information Technology has issued a data protection white paper, with draft regulations. Once enforced, the regulations will provide consumers with protection, will require organisations to have adequate data security practices in place, and specifies the kinds of action that can be taken against companies in case of breaches.

It is always a good idea to use different passwords for various services. You can check if your account was part of a data breach at haveibeenpwned.com. ■